

Profile

M.S. candidate in Chemical Engineering at NTUST, specializing in **electrochemistry**, **ultra-low-temperature electrolytes** and **lithium battery reliability / aging**; co-author on a **Nature Materials** submission.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker

Project lead

Screening methodology

Designed full-stage test protocols, experimental workflows and statistical validation models.

Database & SOP

Established the test database and standardized high-throughput screening procedures (SOP).

Advanced analysis

EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.

Training & tech transfer

Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Materials

Journal submission — battery thermal stability study

Status Under review

Co-author

Responsible for TGA thermal measurement

Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

Experience

2025 – Present

Sustainable Electrochemical Energy Development (SEED) Center, NTUST

Lab Equipment Procurement & Asset Maintenance Manager

- Procurement (>NTD 2M):** Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes, low-temperature baths and TGA platinum pans; also replace instrument consumables.
- Lab safety:** monitor glovebox O₂ / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak.
- Hazardous materials:** handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li-SOCl₂ cells.
- Assembled cells and prepared electrodes in a battery dry room, working in cleanroom garments under clean-area gowning and entry procedures.
- TGA testing service:** Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST — over 600 samples in total, including my own lab's.
- Instrument maintenance:** Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.

Summer 2022

New Taipei City Dept. of Urban Development

Summer Intern, Housing Development Section

- Reviewed rental subsidy eligibility, cross-checking property, income, and insurance records across agencies (tax bureau, land registry, NHI); handled frontline public inquiries and disputes.

2021 – 2024

Tunghai University Student Housing / GEPT

Resident Assistant (2 yrs) / Exam Staff (part-time)

- Oversaw safety and order in a 100-resident dormitory, mediating disputes and tracking facility safety issues with the dorm supervisor; as part-time exam staff, handled test-paper check-in with zero errors.

Conferences & Training

Presenter

2026 TwICHe Annual Meeting

2026/12/05-06 • Feng Chia University

Title: *Weakly Solvating Allyl Ether Electrolytes for Subzero Li || SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability*

Vendor seminar

TA Instruments ARES-G3 launch & applications forum

2026/07/17 • National Taiwan University

Vendor lecture

TA Instruments TGA lecture

2026/06/03 • National Taiwan University

Skills & Instruments

Electrochemistry & batteries

Impedance spectroscopy (EIS)

Battery aging analysis

Reliability analysis

Low-temp electrolytes

Characterization

Thermal analysis (TGA / DSC)

Synchrotron (TLS 20A / TPS 27A / TPS 45A)

Rheology (ARES-G3)

Viscometer & densimeter (DMA 5002)

Process & operations

Lithography (nanoimprint / self-assembly)

Equipment procurement & maintenance

Software

Aspen Plus V14

Aspen Dynamics

Microsoft Office

OriginPro

TA Universal Analysis

TA TRIOS

Education

2025 – 2027 (expected)

National Taiwan University of Science and Technology

M.S. in Chemical Engineering

Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.

2020 – 2024

Tunghai University

B.S. in Chemical & Materials Engineering

Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

EHS Licenses

✓

Dedicated Wastewater and Sewage Treatment Specialist (Class B)

2026 / 06 • certified

✓

Indoor Air Quality Maintenance Management Specialist

2026 / 08 • certified

✓

Waste Disposal Technician (Class A)

2026 / 10 • certified

○

Occupational Safety and Health Management Personnel (Level B)

2026 / 11 • mid-Nov exam scheduled

○

Occupational Safety Management Specialist (Level A)

2027 / 03 • exam scheduled

Instruments

Maintenance

TGA Q500

DSC 2500

Operation

ARES-G3

Anton Paar DMA 5002

Glovebox

Procurement

Glovebox O₂ sensor

DSC sample press

Hydraulic electrode punch

Vacuum pump

Eppendorf Pipette

Low-temperature bath